

CSMProWaterTransport V2.0

Project Aegis-Hydro: Dielectric Water Transportation and Storage Infrastructure Under Heliophysical Load

Version 2.0 — Revised & Expanded | June 2026

Δ Change Log V1.0 -> V2.0

- GFRP pipe: ASTM D2996-2024 updated — approves to 1,500 psi for water service
 - PEX-a: ASTM F876-25 updated standard compliance
 - Aerogel pipe insulation: polyimide-silica hybrid $\lambda=0.010$ W/m.K (upgraded from $\lambda=0.015$)
 - Geopolymer concrete reservoir: RILEM 2025 — 80 MPa, 78% CO2 reduction
 - Water pump: piezoelectric KNbO3-BaTiO3 (lead-free) replaces V1.0 PZT pump
 - Solar Cycle 25 SSN ~137: GIC water infrastructure threat elevated per updated scenario
 - LoRa sensor mesh: 50-node water network monitoring without metallic sensing elements
 - ZrO2 ceramic check valves and gate valves: new 2025 ISO 9283-compliant data
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1. Steel Water Main GIC Failure — V2.0 Quantified

For the LA Aqueduct (442 km, 1.22 m diameter ductile iron main):

At LA Basin amplified field (43 V/km):

$$I_{GIC} = 43 \text{ V/km} \times 442 \text{ km} = 19,086 \text{ V} = 9,363 \text{ A}$$

Cathodic protection override: Standard CP maintains -850 mV pipe-to-soil potential at 5 mA protection current. GIC at 9,363 A creates +42,000 mV pipe-to-soil -> CP completely overwhelmed -> Faradaic corrosion:

$$m_{dissolved} = 10 \text{ g/A} \times 9,363 \text{ A} \times (6/8760) = 64.2 \text{ kg dissolved in 6 hours}$$

— equivalent to 52 m of pipe wall perforated to zero thickness.

2. PEX-a and GFRP Distribution System

2.1 PEX-a Distribution (ASTM F876-25 V2.0)

ASTM F876-25 updates: - UV stabilizer requirements revised for outdoor exposed runs - Chloramine resistance certification pathway clarified - Maximum working pressure at 23°C: 160 psi (unchanged) - V2.0 change: ZrO2 ceramic fittings (replaces brass compression fittings)

2.2 GFRP Transmission Main (ASTM D2996-2024)

For transmission-size water mains (>600 mm diameter): - GFRP filament-wound per ASTM D2996-2024 - Working pressure: 250 psi (typical municipal transmission: 100–180 psi) - Fatigue life: >107 pressure cycles (matching 50-year design life) - Zero Faradaic corrosion -> no cathodic protection system required (saves \$85/m lifetime)

3. Pumping Infrastructure

3.1 Piezoelectric Water Pump (KNbO₃-BaTiO₃ V2.0)

V1.0 specified PZT piezoelectric pump. V2.0 upgrade to lead-free KNbO₃-BaTiO₃: - d₃₃ = 285 pC/N (comparable performance to PZT-5H at 289 pC/N) - T_c = 310°C — prevents depoling in undrained pumping stations that reach high temperatures during post-Carrington fires - RoHS/REACH compliant — no restricted lead compounds

Pump specifications (50 mm nominal bore): - Flow rate: 45 L/min at 8 bar - Power: 180 W piezo input - Zero copper windings — zero GIC coupling - Actuated by compressed nitrogen (manual hand pump for emergency backup)

3.2 Electromagnetic Pump for Conductive Fluids

For applications requiring pumping of treated water with ionic content: Lorentz-force ceramic-lined linear electromagnetic pump: - BFRP outer housing - ZrO₂ ceramic liner (inert to chlorinated water) - Driving coils: alumina-ceramic coated -> isolated from GIC - No moving parts — no bearing wear, no shaft seal

4. Geopolymer Concrete Reservoir (V2.0)

For water storage reservoir construction, standard OPC concrete replaced with geopolymer:

Mix design per m³ (RILEM 2025): - Fly ash: 380 kg (class F) - Metakaolin: 85 kg - GGBS: 115 kg - NaOH activator (8M): 42 L - Na₂SiO₃ solution (Ms=1.8): 95 L - Water/binder: 0.35 - BFRP rebar reinforcement per ACI 440.11-25

Performance: - Compressive strength 28d: 80 MPa (vs 40 MPa OPC) - CO₂ embodied: 113 kg/m³ (vs 505 kg/m³ OPC) — 78% reduction - Sulfate resistance: superior (no calcium aluminate in geopolymer) - Chloride permeability: 800 coulombs (vs 2,000 for OPC) — enhanced durability for coastal locations

5. LoRa Water Network Monitoring (V2.0)

50-node sensor network covering 442 km LA Aqueduct: - Pressure: FBG optical pressure transducers (no metallic sensing elements) - Flow: acoustic flow meters with PEEK-housed transducers - GIC: passive Schumann monitors detect pre-event SR shift (4-6 hr warning)

- Communication: LoRa 915 MHz, 50-node mesh, self-organizing - Power: solar + LiFePO4,
90-day autonomous

End CSMPProWaterTransport V2.0 | Carrington Storm Motors